

מעבדה 1 בתכנון ספרתי מתקדם

מגישים:

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תכנון התוכנית:

```
library IEEE;
use IEEE.STD_LOGIC_1164.All;
use IEEE.STD_LOGIC_UNSIGNED.All;

entity TB_Q3 is
end TB_Q3;

architecture TB_AAA of TB_Q3 is
  component Q3
  port(
    CLK : in std_logic;
    RESET : in std_logic;
    TC : out std_logic
  );
end component;

  signal TB_RESET: std_logic;
  signal TB_CLK: std_logic:= '1';
  signal TB_TC: std_logic;
begin

  DUT : Q3
  PORT map(
    CLK => TB_CLK,
    RESET => TB_RESET,
    TC => TB_TC
  );
  L1: TB_RESET <= '1',
      '0' after 7 MS;
  L2: TB_CLK <= not(TB_CLK) after 5 MS; -- בשביל שהקלט יפלט כל שניה
end TB_AAA;
```

TB:

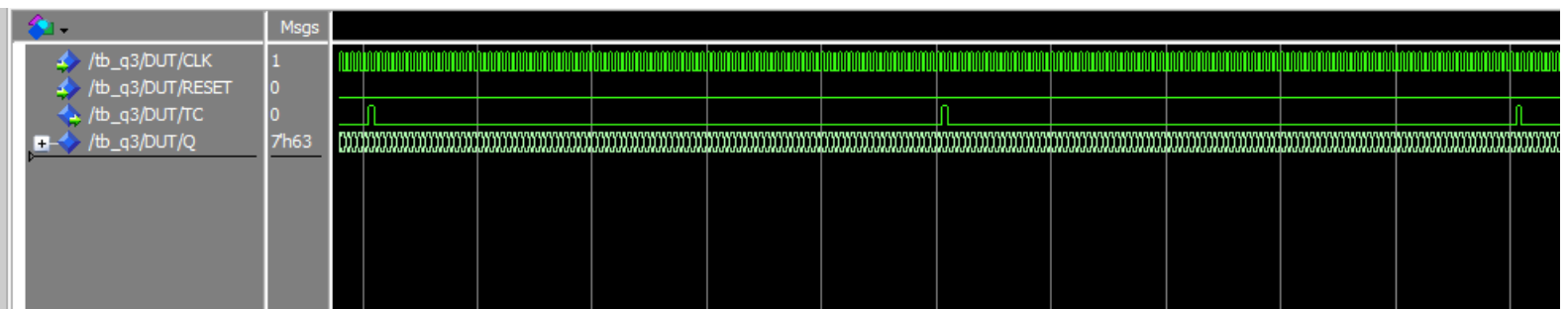
```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity Q3 is
port(
CLK : in std_logic;
RESET : in std_logic;
TC : out std_logic
);
end Q3;

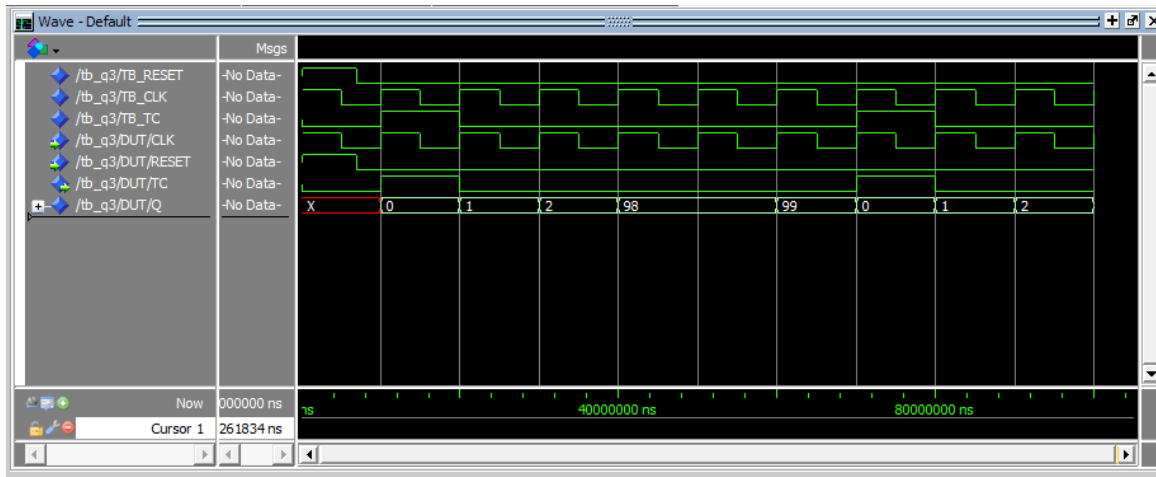
architecture AAA of Q3 is
signal Q : std_logic_VECTOR(6 downto 0);
begin

process
begin
wait until rising_edge(CLK);
if (RESET = '1') then
Q <= (others => '0');
else
if (Q < 99) then
Q <= Q + 1;
else
Q <= (others => '0');
end if;
end if;
end process;
TC <= '1' when (Q = 0) else '0';
end AAA;
```

הסימולציה בלי פקודת ה FORCE



FORCE



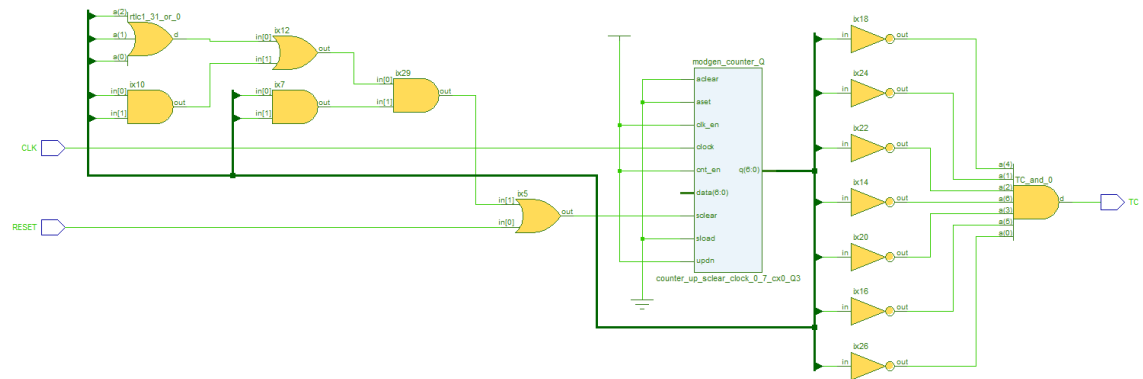
XDC FILE

```

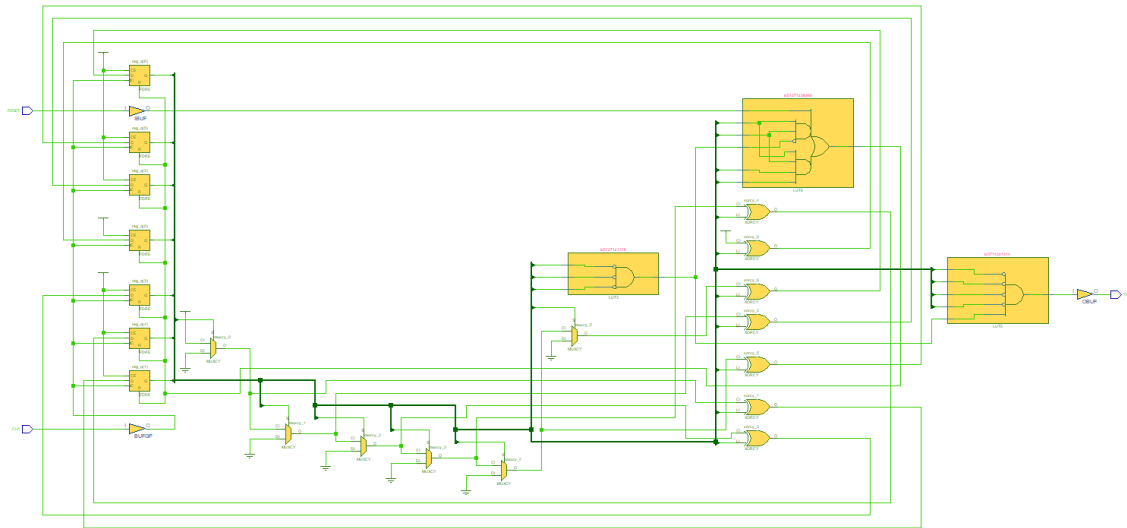
1
2 #####
3 # Clocks #
4 #####
5 create_clock -name CLK -period 10 -waveform {0 5} [get_ports CLK]
6
7 #####
8 # Input Delay #
9 #####
10 set_input_delay 0 -clock CLK -add_delay [get_ports RESET]
11
12 #####
13 # Output Delay #
14 #####
15 set_output_delay 0 -clock CLK -add_delay [get_ports {Q* TC}]
16
17 #####
18 # Pin Locations and Voltages #
19 #####
20 set_property PACKAGE_PIN W5 [get_ports { CLK }];
21 set_property PACKAGE_PIN U18 [get_ports { RESET }];
22 set_property PACKAGE_PIN H1 [get_ports { TC }];
23
24 set_property IOSTANDARD LVCMOS33 [get_ports { CLK }];
25 set_property IOSTANDARD LVCMOS33 [get_ports { RESET }];
26 set_property IOSTANDARD LVCMOS33 [get_ports { TC }];
27
28 #####
29 # Configuration Settings #
30 #####
31 set_property CFGBVS VCCO [current_design]
32 set_property CONFIG_VOLTAGE 3.3 [current_design]
33
34

```

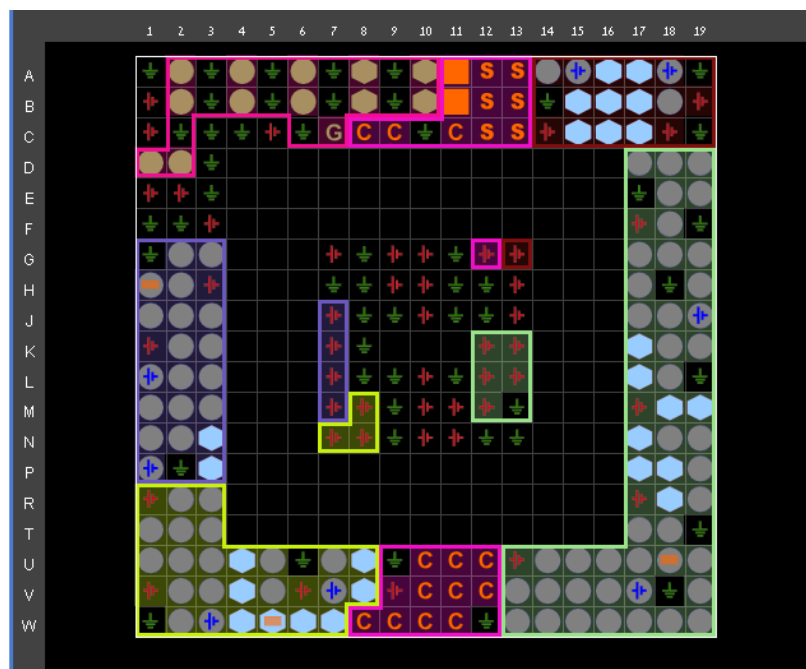
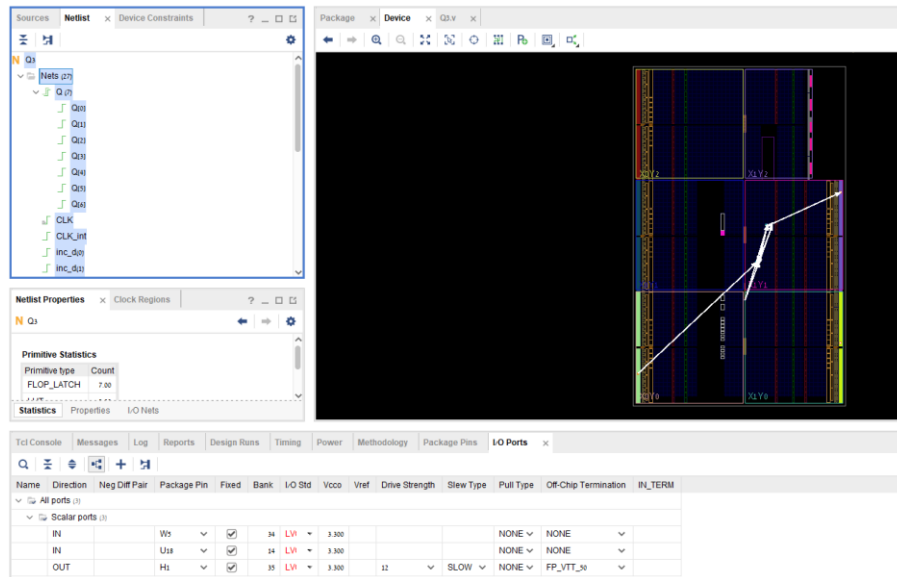
RTL SCHEMATIC



TECH SCHEMATIC



VIVADO DEVICE VIWE AND PIN LOCETION:



.6

10M